



**Programme: B E – Computer Science and Engineering (AI&ML) &
Computer Science and Engineering (Cyber Security)
Internal Assessment – I**

TERM: 3 rd October 2024 to 25 th Jan 2025	COURSE NAME: Data Structures
DATE : 28/11/2024 TIME: 11:00 AM to 12:00 PM	COURSE CODE : CI33/CY33
MAX MARKS: 30	PORTIONS : L1 to L17



Mobile Phones are banned

Instructions to Candidates: Answer any two full questions.

Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Consider a character variable which has been allocated 10 bytes of memory. With suitable C program demonstrate how <i>realloc()</i> function is used to verify and reallocate more bytes without losing data.	L3	CO1	4
b	Write pseudocode for insertion and deletion of an element in a circular queue.	L3	CO2	5
c	Convert the following infix expression to prefix form with appropriate steps using stack representation. E+F-G*H+(I^J)*K/L/M*N+O	L3	CO2	6
2.a	Write a C program to perform insertion and deletion operations on array data structure.	L3	CO1	6
b	Illustrate significance of stack in recursion. Write a C program to print Fibonacci series up to a given number using recursion	L3	CO2	5
c	Evaluate the following postfix expression with appropriate steps using stack representation. 7 5 + 8 4 / * 5 7 * +	L3	CO2	4
3.a	How memory is allocated using malloc() and calloc() functions. Explain with C programming syntax.	L2	CO1	5
b	Consider an online movie ticket reservation system. In this scenario when two customers request same seat booking, which data structure is used to address this issue.	L1	CO2	2
c	Convert the following infix expression to postfix form with appropriate steps using stack representation. A+B-C*D+(E^F)*G/H/I*J+K	L3	CO2	8

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create